RESEARCH
Original Research

Impact of ENHANCED (diEtitiaNs Helping pAtieNts CarE for Diabetes) Telemedicine Randomized Controlled Trial on Diabetes Optimal Care Outcomes in Patients with Type 2 Diabetes

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ABSTRACT

Background Clinical care for type 2 diabetes has improved but remains suboptimal. Collaborative, team-based models that maximize skills of different disciplines may improve care for individuals with diabetes, but few have been tested using rigorous research designs.

Objective To investigate the efficacy of a registered dietitian nutritionist–led telemedicine program compared with that of a control group in terms of diabetes optimal care goals.

Design A randomized controlled trial in which participants were assigned to a control or intervention group.

Participants/setting One hundred eighteen adults with type 2 diabetes (mean age, 60 years; 45% female) participated in the study between April 2016 and December 2017. Participants were recruited from separate primary care clinics in two rural Minnesota communities.

Intervention For those assigned to the intervention, registered dietitian nutritionists used a treatment protocol to initiate and titrate therapies for blood glucose, hypertension, and lipid levels in addition to providing medical nutrition therapy; telemedicine visits supplemented usual care.

Main outcome measures Primary outcomes included composite and individual diabetes optimal care goals: hemoglobin A1c, blood pressure, not using tobacco, and taking a statin and aspirin (as appropriate). Secondary measures included physical activity, breakfast, fruits and vegetables, whole grains, body mass index, low-density lipoprotein, and medication adherence.

Statistical analysis Mixed-model regression was used to examine outcomes between baseline and 1-year follow-up.

Results A modest but significantly greater improvement in the number of diabetes optimal care measures met at follow-up was found in the intervention group (3.7 vs 3.2 in the control group [P=0.017]). Among individual measures, the intervention group had significantly greater medication use, with 2.5 and 2.2 higher odds (compared with the control group) of taking a statin [95% CI, 1.0 to 6.24]) and aspirin [95% CI, 0.90 to 5.19] as appropriate, respectively.

Conclusions ENHANCED (diEtitiaNs Helping pAtieNts CarE for Diabetes) findings suggest that registered dietitian nutritionists following medication treatment protocols can effectively improve care for adults with type 2 diabetes and can serve an instrumental role as part of the health care team in providing evidence-based, patient-centered care.
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MORE THAN 30 MILLION PEOPLE IN THE UNITED States are diagnosed with diabetes, accounting for an estimated 9% of the overall population.[1] Type 2 diabetes accounts for 90% to 95% of diabetes cases, with increasing rates likely the result of rising obesity,[1] as well as age and physical inactivity.[2] In Minnesota, optimal diabetes care is focused on five core measures referred to as the "D5 goals,"[3] which, when met, reduce the risk for medical complications such as cardiovascular disease, renal disease, and retinopathy.[2] These include (1)

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hemoglobin A1c (HbA1c) <8%, (2) blood pressure <140/90 mm Hg, (3) being tobacco free, (4) appropriate statin use, and (5) appropriate aspirin use.[3] These recommendations correlate with the American Diabetes Association Standards of Medical Care.[2]

Recent reviews have indicated general improvements in some diabetes measures,[4,5] but over one-third of adult patients with diabetes in the United States still do not meet glycemic control, blood pressure, or lipid control targets,[4] and 16% smoke.[1] Most diabetes care in the United States is delivered by primary care providers (PCPs). It is well known that PCPs face considerable time constraints in addressing all health issues adequately,[6] which may result in clinical inertia—a well-documented problem that has shown little improvement.[7] The shortage of available PCPs[8] may also be contributing to the inferior rates of diabetes control in places where access to diabetes care and education is limited, such as rural areas.[9]

These factors, as well as frequently changing guidelines (ie, those for HbA1c, blood pressure, and statin use),[2,10] infrequent PCP follow-up, and very low rates of participation in diabetes self-management education the first year after diagnosis (<7%)[11] are barriers to reaching optimal care goals.

Diabetes is a progressive,[12] complex disease, and more innovative models are needed to complement usual diabetes care to keep up with an increasingly burdened health care system. Achieving (and maintaining) diabetes targets may require more extensive, systematic counseling and education, as well as pharmacological therapy, to complement typical clinic-based diabetes care. The HeartBeat Connections telemedicine program was piloted earlier as a platform to complement usual clinical care for cardiovascular disease risk factor control.[13,14] This model was adapted to focus on medical nutrition therapy (MNT), combined with pharmacotherapy, via an approved protocol to address blood pressure and lipid and blood glucose levels for adults with type 2 diabetes. The purpose of this study was to use a randomized controlled trial design to test the effects of a telemedicine intervention provided by a registered dietitian nutritionist (RDN) on type 2 diabetes optimal care goals and other lifestyle outcomes. It was hypothesized that relative to a control group, the intervention group would show greater improvement in the number of diabetes optimal care goals met over 1 year.

RESEARCH SNAPSHOT

Research Question: Does a registered dietitian nutritionist—led telemedicine program utilizing a medication protocol improve diabetes optimal care outcomes for patients with type 2 diabetes?

Key Findings: In this prospective, randomized control trial of 118 adults, modest but significantly greater improvement was observed for the intervention group, relative to the control group, in the number of diabetes optimal care measures met, which increased from a baseline of 3.1 to 3.7 at follow-up (P=0.017). The intervention group had a significantly greater increase in medication use, with 2.5 and 2.2 higher odds (compared with the control group) of taking a statin and aspirin as appropriate, respectively.

METHODS

Design and Participants

ENHANCED (diEtitiaNs Helping pAtieNts CarE for Diabetes) was a randomized controlled trial designed to evaluate the effectiveness of an RDN-led phone coaching intervention that provided MNT and use of a medication therapy protocol. Participants were invited from patient panels in two different health care systems in rural Minnesota, located 60 to 90 miles from the Minneapolis metropolitan area (New Ulm Medical Center and Hutchinson Health). Both facilities include a hospital and a primary care clinic in a single center.

All potentially eligible patients were identified by applying the inclusion and exclusion criteria to their electronic health records (EHRs) at the two study locations. Inclusion criteria were as follows: designated PCP at the Hutchinson or New Ulm clinic, at least one medical encounter at the respective clinic in the previous 2 years, diagnosis of type 2 diabetes by International Classification of Disease 10 codes (ICD 250 and E11),[15] age 40 to 75 years, and meeting three or fewer diabetes optimal care measures (A1c <8%, blood pressure <140/90 mm Hg, not using tobacco, taking a statin and aspirin as appropriate). Medication use was determined based on the presence of an active order. Exclusion criteria were as follows: meeting four or more diabetes optimal care measures, type 1 diabetes, chronic kidney disease, human immunodeficiency virus, institutional residence, major cognitive or language barrier (as determined by study staff), other active end-stage disease (eg, late-stage cancer or pulmonary disease, late-stage congestive heart failure, or a late-stage neuromuscular disorder), actively receiving cancer treatment, pregnancy, and inability to speak/read English. Study procedures were approved by the Quorum Institutional Review Board, and all participants signed an informed consent document. This clinical trial was registered under ClinicalTrials.gov under study identifier NCT02980978.

Recruitment

Recruitment was conducted from March to December 2016 (Figure 1). Patients were initially identified by means of EHR data. (All hospitals and clinics in the Allina Health system use the EpicCare EHR system [Epic Systems Corporation], which Allina has branded Excellian). All data were extracted directly from Allina's custom-built enterprise data warehouse. An introductory letter, signed by each patient's PCP, was mailed to eligible patients. Letters contained a description of the study and an invitation to contact the study team by phone for eligibility screening. Up to three additional outreach attempts were made by phone until a response was received. A screening phone call confirmed study eligibility based on current EHR information and self-reported status. A baseline study enrollment visit was then scheduled for interested and potentially eligible patients. All participants were instructed to fast for 12 hours before the first study visit.

Patients were identified from the initial electronic query of EHR data (Figure 1). One hundred twenty patients from the two study clinics ultimately enrolled between April and December 2016, yielding an enrollment rate of 30%. Data from two participants were excluded (one in the intervention group and one in the control group) from the results because

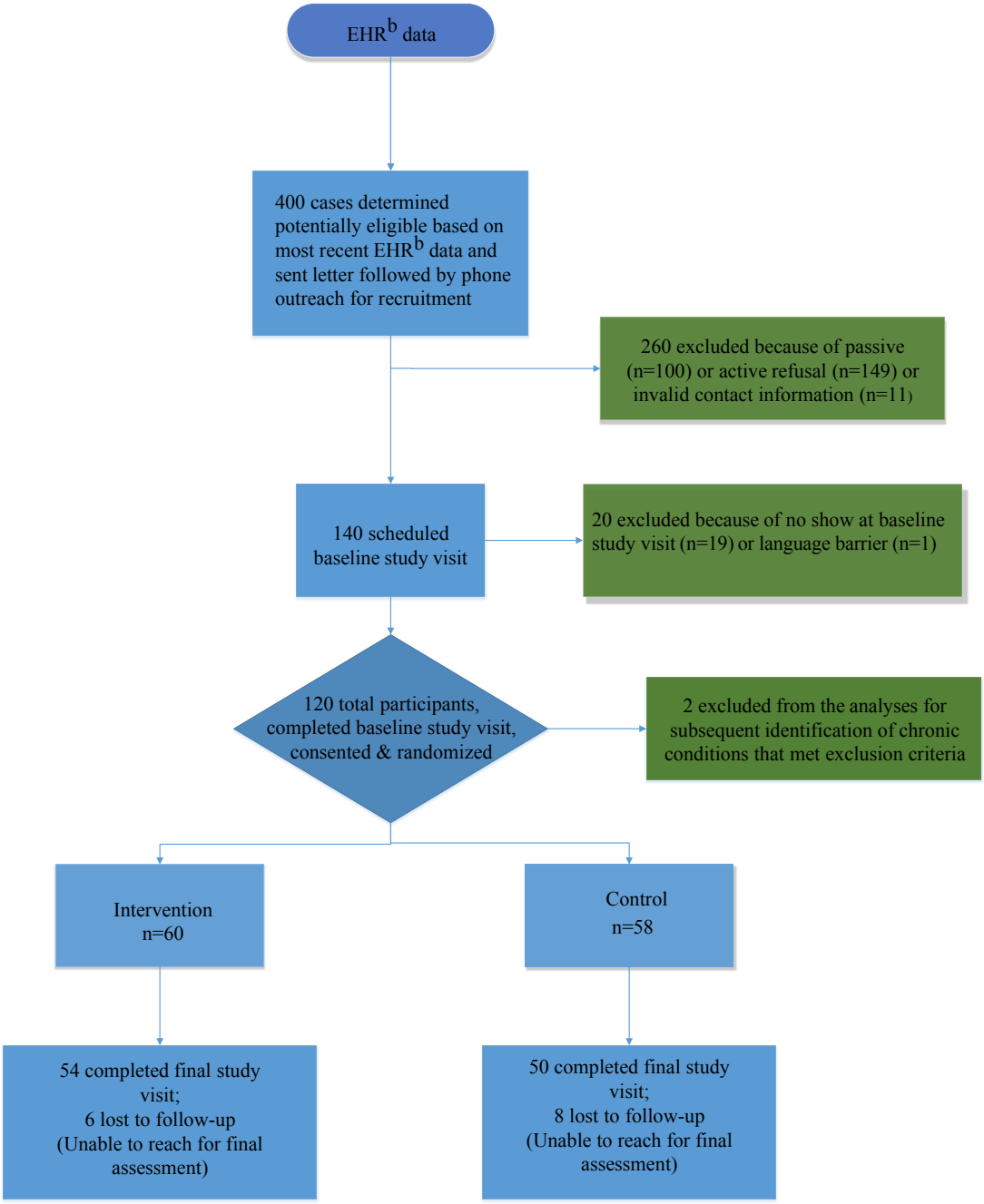

Figure 1. Flow diagram of the ENHANCED[a] randomized controlled trial recruitment process and final study sample of patients with type 2 diabetes from two rural Minnesota primary care clinics. [a]ENHANCED=diEtitiaNs Helping pAtieNts CarE for Diabetes. [b]EHR=electronic health record.

they were found to have severe organ failure shortly after random assignment to groups; thus the final analytical sample included a total of 118 participants. Six participants from the intervention group and eight from the control group were lost to follow-up, resulting in a 12% attrition rate by the 12-month assessment. Intervention participants received an average of 10 telemedicine sessions over the course of the study (range, 3-14). This is the number of monthly sessions offered to participants (excluding the baseline and final assessments); 49 (82%) received eight or more calls, seven (12%) received six or seven calls, and four (7%) received five or fewer calls. Some had more frequent sessions as follow-up to medication initiation or titration. RDN interventionists initiated or titrated medications for 61% of the participants in the intervention group. No adverse events were reported.

Study Components

Figure 2 provides a description of the study components as they apply to all participants and those specific to the control and intervention groups. Measures were collected at two

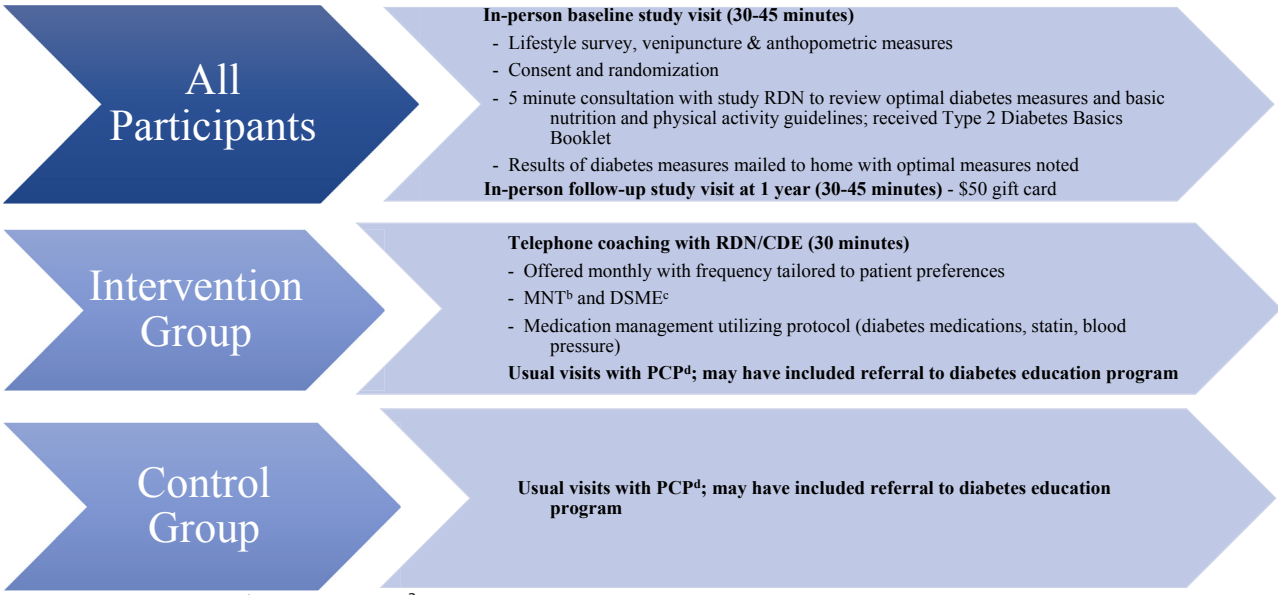

Figure 2. Components of the ENHANCED[a] randomized controlled trial comparing a group that received a telemedicine intervention with a control group in a cohort of adults with type 2 diabetes. [a]ENHANCED=diEtitiaNs Helping pAtieNts CarE for Diabetes. [b]MNT=medical nutrition therapy. [c]DSME=diabetes self-management education. [d]PCP=primary care provider.

in-person study visits at baseline and 1-year follow-up at the patient's primary care clinic. The baseline visit included sociodemographic and lifestyle surveys with self-reported health history and behaviors, venipuncture (for measurement of total cholesterol, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, triglycerides, fasting glucose, and HbA1c), and biometric assessments (blood pressure, height, weight, and waist circumference).

All participants received a copy of their individual results from the baseline visit in the mail with instructions to follow up with their PCP or study coach (if assigned to the intervention group) if any of their results were outside of goal range. Participants were randomly assigned to either the intervention or control group using a computer-generated system that grouped condition assignments into sets of six (three intervention, three control) into envelopes that were opened at the time of assignment. A study staff member who did not perform the randomization filled the envelopes.

Control Group

Those assigned to the control group were told that study staff would reach out to them to schedule a final study visit in 1 year. The 1-year visit was similar to the baseline visit. Participants completed a survey, venipuncture was performed, and biometric measurements were obtained. All study participants who came back for the final study visit received a $50 gift card in appreciation for their participation.

Intervention Group

For those assigned to the phone coaching intervention, initial phone appointments were scheduled for 1 month from the baseline study visit. The intervention was developed using concepts from the Health Belief model[16] and Transtheoretical Model of Behavior Change.[17] It focused on understanding a patient's attitudes and beliefs about his or her health and assessing readiness to change behaviors. Phone coaching intervention approaches included motivational interviewing, diabetes self-management education, identifying and overcoming barriers, goal setting, building self-efficacy, leveraging social support, and coordination with PCPs. Telephone encounters, medication orders, and laboratory orders were documented in the EHR and RDN interventionists communicated directly with participants' PCPs by staff messaging (secure email within the EHR). Medication orders required a cosignature from the PCP. The intervention phone calls included MNT[18] based on the Nutrition Care Process[19] (an initiative of the Academy of Nutrition and Dietetics with the intention of improving the consistency and quality of individualized patient care), as well as the 2015 American Diabetes Association Standards of Medical Care[20] and the National Standards for Diabetes Self-Management Education and Support.[21] Nutrition interventions included meal planning approaches such as the plate method and portion control, carbohydrate counting, and calorie reduction, as well as emphasis on inclusion of adequate fruits/vegetables (five or more servings daily) and healthy fats. Participants were also encouraged to consume whole grains (at least three 1-oz servings per day).[22] These evidence-based nutrition recommendations were individualized for personal preferences (eg, tradition, culture/religion, health beliefs, and economics) and metabolic goals.[23] In addition, RDN interventionists encouraged regular physical activity (at least 150 minutes per week)[20] and medication adjustments as needed for blood pressure, cholesterol (ie, statin), or blood glucose control according to a provider-approved medication protocol for this study.

The stepwise protocol for blood glucose medications started with metformin and included adjustments for insulin for those not meeting blood glucose goals (ie, HbA1c >8%). These

guidelines were designed using the International Diabetes Center Staged Diabetes Management guidelines (5th edition).[24] Use of aspirin was encouraged for those who qualified based on atherosclerotic cardiovascular disease risk score ≥10% who had no history of gastrointestinal bleed and who were not currently receiving anticoagulation therapy.[20,25] Statin therapy was recommended for all individuals in the study who did not have a contraindication (eg, intolerance) according to American Heart Association/American College of Cardiology guidelines.[25] Medications were not provided by the study; all patients' current or new medications were processed per usual billing processes similar to those for a primary care visit. According to the Academy of Nutrition and Dietetics Standards of Practice and Standards of Professional Performance,[26] medication adjustments by RDNs for patients receiving diabetes care depends on education, training, credentialing, and demonstrated competence to practice. For this study, both RDN interventionists (a total of two interventionists for the study) received their certified diabetes educator certification from the National Certification Board for Diabetes Educators.[27] Each interventionist had more than 15 years' experience providing individualized MNT and diabetes self-management education. In Minnesota, the ability to adjust medications was added to state licensure regulations in 2013.[28]

During phone calls, reference was also made to the International Diabetes Center's *Type 2 Diabetes Basics* booklet.[29] Sections of the book were assigned in most cases (eg, "nutrition" in order of participant preference) and discussed during the monthly phone visit. Goals were set collaboratively between the participant and RDN interventionist on completion of each visit. Readiness to change[17] was assessed to inform goals and to follow up on them at the next visit. Intervention calls generally lasted about 30 minutes, with specific call frequency tailored to patient preferences (with more frequent contact after a medication initiation or based on participant availability).

Data Collection and Measures

Primary Outcomes. The primary outcome was the composite number of diabetes optimal care goals met. Five factors were considered based on Minnesota Community Measurement guidelines[3] and included HbA1c <8%, blood pressure <140/90 mm Hg, not using tobacco, taking a statin, and using aspirin as appropriate. Blood pressure was measured after participants had been sitting for 5 minutes. Two blood pressure measures were taken, each 1 minute apart, with an automated sphygmomanometer. The final value was the mean of these two measures. HbA1c was measured by the primary care clinic laboratory with the fasting blood sample collected at each "in person" baseline and final study visit. Aspirin use was measured by means of a single question on the baseline and follow-up surveys that asked participants whether they took an aspirin daily or every other day.[30] People who responded "yes" or who had a documented contraindication to aspirin therapy in the EHR were counted as meeting the optimal care goal. Similarly, statin use was self-reported, and participants who responded "yes" or who had a documented contraindication in the EHR were counted as meeting the optimal care goal. Current tobacco use was measured as a single item ("Do you use tobacco?") with former and never users meeting the optimal care goal. Each of the individual measures was coded as a dichotomous measure with a value of 1 if the individual was meeting the goal and 0 if not. The composite measure for the total number of optimal measures was calculated by adding the values of the five individual measures, resulting in a possible score of 0 to 5 for each participant (Figure 3).

Secondary Outcomes. Other than body mass index (BMI) and LDL level, all secondary outcomes were self-reported at the baseline and follow-up visits via a survey. BMI was calculated by dividing weight (measured with digital scale, in light street clothes with no shoes) in kilograms by height (measured with a stadiometer and no shoes) in meters squared at the first and final visits. BMI was categorized as obese (30.0+), overweight (25.0 to 29.9), or not obese/overweight (≤24.9) per national definitions.[31] LDL level was determined by the primary care clinic laboratory with the fasting blood sample collected at each study visit. Medication use was assessed by questions that asked whether the participant had a prescription for blood pressure, cholesterol, or diabetes. If a participant answered "yes" to one of these, medication adherence was measured separately for cholesterol, blood pressure, and diabetes medication using the four-item Morisky scale.[32] Physical activity was reported using a screening question developed for the Behavioral Risk Factor Surveillance System.[33] This four-part question captures both vigorous and moderate activity in number of minutes per week. Sufficient physical activity was defined as at least 150 minutes of moderate intensity activities per week. Fruit and vegetable intake was measured with two questions indicating total daily fruit and total daily vegetable servings (cups per day). The question was modified from the American Heart Association Simple 7.[34] Whole grains were measured by asking how many servings were eaten each day based on serving sizes listed. The question was modified from the American Heart Association Simple 7[34] and based on the recommendations of the 2015-2020 Dietary Guidelines for Americans to eat more whole grains and fewer refined grains.[35] Breakfast was assessed using a question from the Stanford Patient Education Research Center diabetes questionnaire,[36] which asks about the number of days breakfast was eaten in the past week.[37]

Patient Characteristics and Health History. Other than age, sex, and clinic assignment, all patient baseline characteristics and health history measures were self-reported on the baseline survey. Race/ethnicity was one question. Diabetes-related questions addressed duration of condition, insulin use, and previous formal diabetes education. Health history was assessed by asking whether the patient had ever been told by a doctor that he or she had any conditions listed in the response options.

Statistical Analyses

The target sample size was at least 50 participants in each group. This was based on achieving a moderate Cohen's W effect size of 0.28 between groups regarding the proportion of participants who achieved improvement in at least one diabetes optimal care goal over 1 year. This would roughly translate into about 33% of the intervention group

Type	Measures	Description
Biometrics (laboratory)	HbA1c[ab] LDL[c] cholesterol	Venipuncture after request for 12-hour fast; standard clinic medical laboratory blood collection and processing procedures
Anthropometrics (staff measured)	Blood pressure[b]	Staff measured; participants sat for 5 min before three pressures were taken 1 min apart using automated sphygmomanometers (mean of last two measures recorded); if <140/90 mm Hg, individuals were considered to meet the optimal care goal, and if ≥140/90 mm Hg, to not meet it.
	Body mass index	Staff measured; calculated by dividing weight (measured with digital scale, in light street clothes without shoes) in kilograms by height (measured with a stadiometer and no shoes) in meters squared; categorized as obese (30.0+), overweight (25.0−29.9), or not obese/overweight (≤24.9) per national guidelines.[31]
Medication (self-report survey)	Aspirin use[b]	Self-reported; single item indicating use of aspirin every day or every other day.[30] People who responded "yes" or who had a documented contraindication to aspirin in the electronic health record were counted as meeting the optimal care goal.
	Statin use[b]	Self-reported; single item indicating statin use.[50] People who responded "yes" or who had a documented contraindication to a statin in the electronic health record were counted as meeting the optimal care goal.
	Insulin use	Self-reported; single item indicating insulin use
	Medication adherence (cholesterol, diabetes, blood pressure)	Self-reported; 15 items; the Morisky Medication Adherence Scale[32] is a generic self-reported, medication-taking behavior scale in which the specific health issue is inserted for the "health concern." It consists of four items with a scoring scheme of "yes"=0 and "no"=1. The items are summed to give a range of scores from 0 to 4. High adherence is categorized as a score of 4 or higher.
Lifestyle (self-report survey)	Tobacco use[b]	Self-reported; single item indicating current user, former user, or never used tobacco; former and never users were considered to meet the optimal care goal.
	Physical activity	Self-reported; screener developed for the Behavioral Risk Factor Surveillance System[33]; four items; moderate and vigorous physical activity reported in days per week and minutes per day; reported in moderate equivalent units by doubling the vigorous minutes per week and adding them to the reported moderate minutes per week; categorized as sufficient (>150) or insufficient (<150) per national guidelines.
	Fruit, vegetable, and whole-grain intake	Self-reported; three items modified from American Heart Association Simple 7[34]; daily intake of fruits (cups/d), vegetables (cups/d), and whole grains[35] (servings/d)
	Breakfast	Self-reported; single item; frequency; individuals were considered "breakfast eaters" if they reported eating breakfast 4-7 days/wk.[36]
Diabetes-related (self-report survey)	Duration of diabetes	Self-reported; single item; years since diabetes diagnosis
	Diabetes education class	Self-reported; single item; ever taken a course or class on diabetes management (yes/no/don't know)[51]

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Figure 3. Primary and secondary outcome measures of the ENHANCED (diEtitiaNs Helping pAtieNts CarE for Diabetes) randomized controlled trial comparing a group that received a telemedicine intervention with a control group in a cohort of adults with type 2 diabetes.

Type	Measures	Description
Health history (self-report survey)	Other conditions	Self-reported; selected from list of health conditions other than diabetes. Participants were asked to check if a provider had ever told them they had any of the following conditions: gestational diabetes, preeclampsia, polycystic ovarian syndrome, high blood pressure, high cholesterol, depression, stroke or TIA[d], heart attack, cardiac stent, coronary bypass surgery, coronary artery disease, congestive heart failure, kidney disease, neuropathy; none of the above and "don't know/choose not to answer" were also available.

[a]HbA1c=hemoglobin A1c.
[b]Primary outcome measures of ENHANCED.
[c]LDL=low-density lipoprotein.
[d]TIA=transient ischemic attack.

Figure 3. *(continued)* Primary and secondary outcome measures of the ENHANCED (diEtitiaNs Helping pAtieNts CarE for Diabetes) randomized controlled trial comparing a group that received a telemedicine intervention with a control group in a cohort of adults with type 2 diabetes.

experiencing improvement in at least one optimal care goal versus 10% of control participants (under assumptions of 80% power, an α level of .05, and a 1 degree of freedom test). Baseline values for participants in the intervention and control groups were compared by using χ^2 tests for categorical measures and *t* tests for continuous measures. For comparison of change over time for both primary and secondary outcomes, mixed-effects longitudinal models were used with an intent-to-treat framework, which included fixed effects for time and intervention group and a random subject-specific effect. Linear mixed-effects models were used for continuous measures and mixed effects logistic regression for binary variables. For both models, an unstructured working correlation was specified. A time-by-group interaction term was used to determine whether the changes over time differed between intervention and control groups for all outcomes. Model-based coefficients were used to construct baseline and follow-up point estimates (means, proportions, and robust standard errors) for the intervention and control groups. As a subanalysis, *t* tests and $\chi 2$ tests were conducted to compare follow-up values among the subset of participants who were not lost to follow-up. All analyses were conducted using Stata version 15[38]; a two-sided α value of .05 was considered statistically significant.

RESULTS

The average age of participants was 60 years, and 45% were female. Forty-one percent of participants were from the New Ulm clinic, and 59% were from Hutchinson. There were no statistically significant differences between the intervention and control groups at baseline with regard to demographics, BMI, comorbidities, HbA1c, or optimal diabetes measures met (Table 1). Of note, patients were initially identified by selecting those who met eligibility criteria according to EHR data. Those who agreed to participate provided consent and were randomly assigned to the intervention or control group at the baseline assessment visit. Optimal care goals were reviewed with each patient and reassessed at the time of the baseline visit by RDN interventionists. Despite being identified as having three or fewer optimal care goals at baseline according to EHR data, 31% of the sample were found to be meeting four to five optimal care goals at the time of the baseline visit, with no significant difference in allocation of these individuals between the baseline and control groups (P=0.53).

The group-by-time interaction in which an intent-to-treat methodology was used for the total number of diabetes optimal measures achieved was significant (P=0.017), indicating that those in the intervention group showed significantly greater improvement, relative to those in the control group, in the composite endpoint of all diabetes optimal care goals between baseline and 1-year follow-up (Table 2). A borderline significant group-by-time interaction was also observed for statin use (P=0.050). Compared with participants in the control group, those who did not use statins in the intervention group had more than twice the odds of initiating statin use by the 1-year follow-up visit; however, this did not result in a significant difference in LDL reduction between the two groups.

No other significant group-by-time interactions were observed, indicating that the magnitude of change for most individual diabetes measures was somewhat similar in both groups. Within-group changes from baseline to follow-up for the intervention group showed significant improvement for the composite optimal diabetes care measure and statin and aspirin use but no difference in HbA1c level, blood pressure, or tobacco use. The control group had significant improvement between baseline and follow-up for the composite optimal diabetes care measure, HbA1c level and aspirin use but no significant change in blood pressure or statin or

Table 1. Baseline characteristics for ENHANCED[a] randomized controlled trial enrollees (n=118), which compares a group that received a telemedicine intervention with a control group in a cohort of adults with type 2 diabetes

	Intervention (n=60)	Control (n=58)	*P* value
	←mean±SD→		
Age, y	59.8±10.20	60.0±8.66	0.894
	←%→		
40-49	18.3	13.8	0.399
50-59	31.7	27.6	
60-69	30.0	44.8	
70-75	20.0	13.8	
Sex			
Female	46.7	43.1	0.697
Male	53.3	56.9	
Race/ethnicity			
African American	3.3	0.0	0.150
Asian	1.7	1.7	
Pacific Islander/Native Hawaiian	0.0	0.0	
Hispanic	0.0	3.5	
American Indian/Alaska Native	0.0	0.0	
White	93.3	89.7	
Multiracial	0.0	5.2	
Unknown	1.7	0.0	
Education			
High school or less	28.3	31.0	0.535
Some college, technical or Associate's Degree	50.0	55.2	
Bachelor's degree or higher	21.7	13.8	
Clinic			
New Ulm, MN	41.7	41.4	0.975
Hutchinson, MN	58.3	58.6	
Duration of diabetes diagnosis			
< 1 y	8.3	10.3	0.376
1-3 y	13.3	19.0	
3-5 y	10.0	15.5	
5-10 y	25.0	29.3	
> 10 y	40.0	25.9	
Don't know	3.3	0.0	
Insulin use for diabetes management			
Yes	45.0	30.4	0.104
No	55.0	69.6	
Self-reported health history			
Reproductive related conditions[b]	5.0	5.2	0.966
High blood pressure	70.0	67.2	0.747
High cholesterol	48.3	50.0	0.856

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Table 1. Baseline characteristics for ENHANCED[a] randomized controlled trial enrollees (n=118), which compares a group that received a telemedicine intervention with a control group in a cohort of adults with type 2 diabetes (*continued*)

	Intervention (n=60)	Control (n=58)	*P* value
Depression	33.3	22.4	0.186
Cardiovascular disease[c]	11.7	10.3	0.819
Congestive heart failure	3.3	0.0	0.161
Kidney disease	11.7	3.5	0.093
Neuropathy	13.3	13.8	0.942
Retinopathy	5.0	8.6	0.434
Ever attended a diabetes education class			
Yes	55.0	51.7	0.518
No	45.0	44.8	
Don't know	0.0	3.5	
	←mean±SD→		
Total optimal diabetes measures met	3.1±0.96	2.9±0.96	0.129
Hemoglobin A1c	8.1±1.55	8.3±1.66	0.529
Low-density lipoprotein cholesterol	79.5±34.66	83.4±29.86	0.517
Body mass index categories[31]	37.8±9.80	36.2±6.21	0.281
	←%→		
Healthy weight (<24.9)	5.0	3.5	0.862
Overweight (25-29.9)	8.3	10.3	
Obese (30+)	86.7	86.2	

[a]ENHANCED=diEtitiaNs Helping pAtieNts CarE for Diabetes.
[b]Reproductive related conditions includes gestational diabetes, preeclampsia, and polycystic ovarian syndrome.
[c]Cardiovascular disease includes stroke or transient ischemic attack, heart attack, cardiac stent, coronary bypass surgery, coronary artery disease.

tobacco use. For the purpose of exploring the possible influence of participant attrition on these findings, a sensitivity analysis was conducted that excluded the 14 patients with missing 12-month follow-up data. Crude *t* tests and $\chi 2$ tests were used to compare the observed follow-up values between the intervention and control groups (Table 3). Results from this analysis were nearly identical to those from the intent-to-treat, mixed models.

Secondary outcomes assessed nutrition, physical activity, and medication adherence, as well as BMI, and LDL and HbA1c levels (Table 2). Significant group-by-time interactions indicate greater improvements for intervention participants in consumption of fruits (P=0.011) and whole grains (P=0.005), as well as diabetes medication adherence (P=0.014). No other secondary measures differed significantly between groups, including mean HbA1c level, which showed similar improvement between baseline and follow-up in both groups. As with the primary outcomes, a crude sensitivity analysis, which excluded participants who were unavailable at follow-up, was run on these secondary outcomes. Again, results (not shown) were not different from those of the intent-to-treat, mixed models.

DISCUSSION

This was the first known trial of a telemedicine intervention delivered by RDNs using medication therapy management protocols to manage optimal diabetes care measures for patients with type 2 diabetes. Compared with a control group receiving usual care, participants who received the RDN-led telemedicine intervention had significantly greater improvement in the total number of diabetes optimal care measures met. The magnitude of this improvement in the intervention group was fairly modest, averaging about 0.6 more diabetes care goals met at follow-up relative to baseline, but it was about twice the "rate" of improvement observed among participants who only received usual clinical care. Additionally, the intervention group had significantly greater improvements in fruit and whole grain consumption, as well as diabetes medication adherence. Promoting healthy

Table 2. Model-based optimal diabetes measure outcomes among all adult participants with type 2 diabetes in the ENHANCED[a] trial (n=118)

Primary outcomes	Intervention (n=60) Baseline	Intervention (n=60) Follow-up	Control (n=58) Baseline	Control (n=58) Follow-up	Interaction *P* value[b]	Mean difference[c]	SE[d]
	←mean (SE)→						
Total no. of optimal diabetes measures achieved	3.1 (0.12)	3.7 (0.13)[e]	2.9 (0.13)	3.2 (0.13)[e]	0.017	0.43	0.18
Individual measures met	←% (SE)→					**OR**[cf]	**CI**
Blood pressure < 140/90 mm Hg	33.3 (0.06)	44.4 (0.07)	46.6 (0.07)	38.2 (0.07)	0.538	1.28	[0.59, 2.78]
Statin use (or contraindicated)	60.0 (0.06)	82.7 (0.05)[e]	56.9 (0.07)	65.7 (0.07)	0.050	2.50	[1.00, 6.24]
Hemoglobin A1c < 8%	50.0 (0.06)	64.8 (0.07)	48.3 (0.07)	68.3 (0.06)[e]	0.702	0.85	[0.38, 1.92]
Not using tobacco	78.3 (0.05)	78.7 (0.05)	77.6 (0.05)	79.6 (0.05)	0.903	0.95	[0.38, 2.32]
Taking daily aspirin (or contraindicated)	63.3 (0.06)	81.7 (0.05)[e]	56.9 (0.07)	67.4 (0.06)[e]	0.084	2.16	[0.90, 5.19]
Secondary outcomes							
	←% (SE)→						
Physical activity (150 min of moderate activity/wk)	53.3 (0.06)	58.0 (0.07)	46.6 (0.07)	52.2 (0.07)	0.550	1.26	[0.59, 2.73]
Breakfast (4-7 d/ wk)	73.3 (0.06)	73.6 (0.06)	67.2 (0.06)	64.4 (0.07)	0.299	1.55	[0.68, 3.52]
						Mean difference	**SE**
	←mean (SE)→						
Fruit (cups/d)	1.3 (0.13)	1.6 (0.14)[e]	1.1 (0.14)	1.1 (0.11)	0.011	0.45	0.18
Vegetables (cups/d)	1.7 (0.17)	1.9 (0.14)	1.5 (0.15)	1.6 (0.17)	0.200	0.28	0.22
Whole grains (servings/d)	1.6 (0.15)	1.9 (0.16)	1.7 (0.17)	1.3 (0.13)[e]	0.005	0.57	0.20
Hemoglobin A1c	8.1 (0.20)	7.4 (0.15)[d]	8.3 (0.22)	7.7 (0.20)[e]	0.388	−0.21	0.25
Low-density lipoprotein cholesterol	79.5 (4.46)	66.4 (5.22)[d]	83.4 (3.90)	76.7 (5.31)	0.164	−10.36	7.44
Body mass index	37.8 (1.26)	37.9 (1.32)	36.2 (0.81)	35.7 (0.83)	0.162	2.19	1.56
Medication adherence changes among those taking medication at baseline						**OR**	**CI**
Cholesterol medication use	(n=34)	(n=34)	(n=33)	(n=29)			
	←% (SE)→						
High adherence	73.9 (0.07)	82.6 (0.06)	59.0 (0.08)	83.0 (0.07)[e]	0.959	0.97	[0.29, 3.29]
Diabetes medication use	(n=42)	(n=36)	(n=46)	(n=37)			
High adherence	57.1 (0.08)	75.3 (0.07)[e]	42.6 (0.07)	47.4 (0.08)	0.014	3.38	[1.28, 8.96]
Blood pressure medication use	(n=40)	(n=40)	(n=42)	(n=37)			
High adherence	78.6 (0.06)	90.9 (0.04)	75.5 (0.07)	76.5 (0.07)	0.088	3.05	[0.85, 11.02]

[a]ENHANCED=diEtitiaNs Helping pAtieNts CarE for Diabetes.
[b]*P* values correspond to the group-by-time interaction from the model comparing intervention group with control groups on baseline-to-follow-up change. This analysis was conducted under an intent-to-treat framework.
[c]Odds ratio and mean differences are comparing the point estimates for the intervention and control groups at follow-up.
[d]SE=standard error.
[e]Values indicate a significant ($P<0.05$) change from baseline to follow-up within either the intervention or control group.
[f]OR=odds ratio.

Table 3. Primary outcomes at 1-year follow-up among adults with type 2 diabetes who participated in the ENHANCED[a] trial and had outcome data at follow-up (n=104)[b]

Primary outcomes	Intervention (n=54)	Control (n=50)	*P* value
Total no. of optimal diabetes measures achieved, mean±SD[c]	3.7±0.93	3.3±0.92	0.028
Individual measures met, %			
Blood pressure <140/90 mm Hg	44.4	40.0	0.647
Statin use (or contraindicated)	81.5	64.0	0.045
Hemoglobin A1c <8%	64.8	70.0	0.573
Not using tobacco	81.5	80.0	0.848
Taking daily aspirin (or contraindicated)	83.3	68.0	0.068

[a]ENHANCED=diEtitiaNs Helping pAtieNts CarE for Diabetes.
[b]Baseline and follow-up data were available for 104 participants. Fourteen participants were lost to follow-up and excluded from this subanalysis.
[c]SD=standard deviation.

eating, regular physical activity, and other lifestyle behaviors is well established as part of the RDN scope of practice,[39] whereas RDN involvement in some other aspects of diabetes care, such as medication management, tends to be more limited. The findings from this study generally support the limited, but growing, body of evidence that RDN-led interventions that include prescribing rights can enhance patient care outcomes.[13,40,41]

Although general improvements were experienced in most diabetes care goals, more favorably so in the intervention group, differences, specifically in HbA1c control, between the intervention and control groups at 1 year were not significant. Previous research has indicated that MNT is most effective for those with higher baseline HbA1c values,[23] and both groups were fairly close to goal at study initiation. Many patients may have benefited from a more aggressive goal (ie, <7%), but medication adjustments were not allowed according to the protocol with HbA1c <8%. Even so, both groups ended the study with only 65% of the intervention group and 70% of the control group meeting the Minnesota Community Measurement goal of <8%.[3] It is also important to note that 40% of the intervention group and 26% of the control group had had a diagnosis of diabetes for more than 10 years. Studies have revealed that the largest reductions in HbA1c occur soon after diagnosis, and because of the progressive nature of the diabetes, HbA1c increases about 1% every 2 years, requiring more vigorous interventions.[12]

A potentially contaminating factor was that one of the study sites implemented a new initiative during the study whereby clinic nurses called all patients with an HbA1c >8% to consider medication adjustments. In addition, the study provided robust education and communication to on-board sites. PCPs were aware of the study goals and may have changed practice as a result. Affordability of medications also emerged as a concern often mentioned in RDN chart notes in the intervention group (and was presumably a similar concern in the control group), which may have made additional medication adjustments (eg, insulin, non-insulin injectable diabetes medications) difficult.[42] Many patients struggled with medication costs and often limited their doses or skipped doses altogether to extend their medication supply over time.

Achieving optimal diabetes care requires a broader focus compared with glycemic control alone.[43] Diabetes is associated with significantly elevated risk of cardiovascular disease, and overall health can be significantly improved by managing multiple risk factors at the same time.[25] If lifestyle changes are combined with appropriate medication use, cardiovascular disease events may be reduced by as much as 53%.[44] The group-by-time analysis revealed that those assigned to the intervention group were more than twice as likely to be taking a statin or blood pressure medication as appropriate at 1 year compared with those in the control group. The medication protocol in this study, which allowed initiation/titration for statin, blood pressure, and diabetes medications, builds on a small body of research in which medication adjustments by RDNs were examined. Robinson and colleagues[40] demonstrated that RDNs who followed a lipid-titration algorithm were able to double the number of patients meeting lipid goals. Worth and Davies[41] found that RDNs who recommended, monitored, and titrated lipid-lowering therapies were able to achieve significant improvements in both total cholesterol and triglyceride levels. In the Remedies 4D trial, patients with type 2 diabetes from 15 primary care practices (n=240) were randomly assigned to usual care or an intervention that utilized preapproved medication protocols. In the intervention, certified diabetes educators (registered nurses by background) adjusted medications to improve HbA1c, blood pressure, and LDL cholesterol levels. Significant improvement in HbA1c was observed at 1 year, but no differences in blood pressure or LDL cholesterol were found between the intervention and control groups.[45] These team-based, coordinated models of care are underscored in the Community Preventive Services Task Force's recommendations for type 2 diabetes.[46]

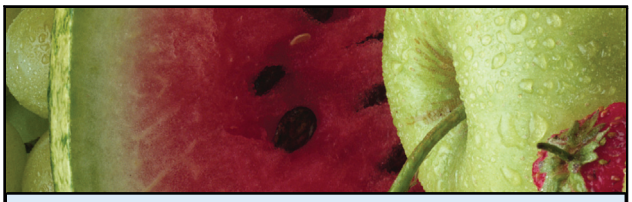
A comprehensive approach to diabetes management with extensive lifestyle counseling, patient education, and pharmacological therapy—and frequent follow-up to address

challenges as they arise—is needed to manage comorbidities and prevent complications.[43] The National Standards of Diabetes Self-Management Education and Support emphasize the importance of an interprofessional team approach including RDNs to improve access, clinical outcomes, and cost-effectiveness, while keeping the individual with diabetes as the center of the care team.[21] According to a recent Academy of Nutrition and Dietetics position statement,[47] RDNs are well suited to add value to clinical care of patients with diabetes by offering comprehensive holistic care, including medication management and MNT. Lifestyle behaviors including nutrition, physical activity, and medication adherence were addressed during RDN intervention calls (in addition to the primary diabetes optimal care outcome measures). Coaching around these topics was individualized based on participant interest and readiness to change.[17] A recent systematic review and meta-analysis compared individualized nutrition therapy by an RDN to that of dietary advice given by other providers and demonstrated a greater effect on diabetes outcomes (HbA1c, weight, and cholesterol) when provided by an RDN.[48] ENHANCED study findings were consistent with prior research in that individualized behavior change coaching by an RDN resulted in improvement in overall diabetes care measures met, diabetes medication adherence, and some lifestyle measures (eg, intake of fruit and whole grains) but no differences in terms of HbA1c or weight.

This study's findings were strengthened by the randomized controlled trial design and multisite implementation over a full year. Additionally, the method of analysis using the mixed-model regression incorporates all data available for participants, regardless of dropout. Although the study sample size was modest, attrition was low, and all intervention group participants received at least some level of telemedicine follow-up. A limitation was inclusion of some participants who were found to be meeting four or five diabetes measures at the time of baseline measures. Additionally, some proactive phone medication titration was conducted by registered nurses independently of this study at one clinic site. The control group also received the International Diabetes Center's *Type 2 Diabetes Basics* booklet[29] (and were advised to follow up with their PCPs if goals were out of range). These features could have collectively muted the impact of the intervention and biased conclusions toward the null. Also, the participants had fairly complex disease complications, often requiring multiple combinations of medications to manage HbA1c and blood pressure. The medication protocol remained unchanged throughout the study but would likely benefit from updating to include new therapies as they become available. (The Food and Drug Administration approved more than 25 new diabetes drugs in the last decade).[49]

CONCLUSION

In this randomized controlled trial, modest but significantly greater improvement was observed for the RDN-led intervention group, relative to the control group, in the number of diabetes optimal care measures met. If the role of the RDN is expanded to include medication management, patients may be more likely to achieve their diabetes optimal care goals. Nutrition coaching via telephone expands the RDN role further by addressing the barrier of geographic inaccessibility and patient and provider time constraints. This study more directly shared the responsibility of both diabetes care and diabetes self-management education between PCPs and RDNs. It also tested an expanded, nontraditional role of RDNs in this context, whereby they effectively delivered both nutrition and medication therapies remotely via telephone, but in full complement to and coordination with clinic-based primary care.



PRACTICE IMPLICATIONS

Health care systems, especially those with provider shortages, could use what has been learned from this study and consider preapproved medication protocols for RDNs to use as part of the expanded care coordination team. By documenting medication adjustments in the EHR, the RDN is in direct communication with a PCP who can efficiently cosign medication orders. The present study might be replicated in other communities, perhaps on a larger scale, to further investigate RDNs' impact on diabetes care outcomes. Collectively, this research could be used to prompt a Medicare rule change that would allow RDNs to direct bill Part B for the new remote patient monitoring benefit, direct bill for the long-term care management benefit, allow MNT coverage for the diagnoses of hypertension and hyperlipidemia, or possibly prompt a change that would allow RDNs to be "eligible clinicians" in the new Medicare Quality Payment Program. Future research, including cost analyses and outcomes monitoring, may help build evidence to support the RDN's role in care coordination and advanced diabetes management.

References

1. Centers for Disease Control and Prevention. National Diabetes Statistics Report, 2017. https://www.cdc.gov/diabetes/data/statistics-report/index.html. Published 2017. Accessed December 12, 2017.
2. American Diabetes Association. Standards of medical care in diabetes 2018. *Diabetes Care*. 2018;41(suppl 1):S1-S172.
3. The D5: Five goals for living well with diabetes. www.mnhealthscores.org. http://mncm.org/wp-content/uploads/2013/04/MNCM_D5_11x14 Poster_FNL2.pdf. Published 2014. Accessed May 10, 2018.
4. Ali MK, Bullard KM, Saaddine JB, Cowie CC, Imperatore G, Gregg EW. Achievement of goals in U.S. diabetes care, 1999-2010. *N Engl J Med*. 2013;368(17):1613-1624.
5. Casagrande SS, Fradkin JE, Saydah SH, Rust KF, Cowie CC. The prevalence of meeting A1c, blood pressure, and LDL goals among people with diabetes, 1988-2010. *Diabetes Care*. 2013;36(8):2271-2279.
6. Konrad TR, Link CL, Shackelton RJ, et al. It's about time: Physicians' perceptions of time constraints in primary care medical practice in three national healthcare systems. *Med Care*. 2010;48(2):95-100.
7. Strain WD, Blüher M, Paldánius P. Clinical inertia in individualising care for diabetes: Is there time to do more in type 2 diabetes? *Diabetes Ther*. 2014;5(2):347-354.
8. Petterson SM, Liaw WR, Phillips RL, Rabin DL, Meyers DS, Bazemore AW. Projecting US primary care physician workforce needs: 2010-2025. *Ann Fam Med*. 2012;10(6):503-510.

9. Purnell TS, Calhoun EA, Golden SH, et al. Achieving health equity: Closing the gaps in health care disparities, interventions, and research. *Health Aff*. 2016;35(8):1410-1415.
10. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/ APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults. *J Am Coll Cardiol*. 2018;71(19):e127-e248.
11. Li R, Shrestha SS, Lipman R, Burrows N, Kolb LE, Rutledge S. Diabetes self-management education and training among privately insured persons with newly diagnosed diabetes — United States, 2011-2012. *Morbidity and Mortality Weekly Report (MMWR)*. https://www.cdc.gov/mmwr/pdf/wk/mm6346.pdf. Published 2014. Accessed November 5, 2018.
12. Fonseca VA. Defining and characterizing the progression of type 2 diabetes. *Diabetes Care*. 2009;32(suppl 2):S151-S156.
13. Benson GA, Sidebottom A, VanWormer JJ, Boucher JL, Stephens C, Krikava J. HeartBeat Connections: A rural community of solution for cardiovascular health. *J Am Board Fam Med*. 2013;26(3):299-310.
14. Benson GA, Sidebottom A, Sillah A, et al. Reach and effectiveness of the HeartBeat Connections telemedicine pilot program. *J Telemed Telecare*. 2018;24(3):216-223.
15. Dugan J, Shubrook J. *International Classification of Diseases*, 10th Revision, Coding for Diabetes. *Clin Diabetes*. 2017;35(4):232-238.
16. Becker MH. The health belief model and personal health behavior. *In: Health Education Mongraphs*. 1974;2:324-508.
17. Prochaska JO, Velicer WF. The transtheoretical model of health behavior change. *Am J Health Promot*. 1997;12(1):38-48.
18. Evert AB, Boucher JL, Cypress M, et al. Nutrition therapy recommendations for the management of adults with diabetes. *Diabetes Care*. 2014;37(suppl 1):120-143.
19. Swan WI, Vivanti A, Hakel-Smith NA, et al. Nutrition care process and model update: Toward realizing people-centered care and outcomes management. *J Acad Nutr Diet*. 2017;117(12):2003-2014.
20. American Diabetes Association. Standards of medical care in diabetes—2015. *Diabetes Care*. 2015;38(Jan):S1-S2.
21. Beck J, Greenwood DA, Blanton L, et al. 2017 National standards for diabetes self-management education and support. *Diabetes Care*. 2017;40(10):1409-1419.
22. He M, Van Dam RM, Rimm E, Hu FB, Qi L. Whole-grain, cereal fiber, bran, and germ intake and the risks of all-cause and cardiovascular disease-specific mortality among women with type 2 diabetes mellitus. *Circulation*. 2010;121(20):2162-2168.
23. Franz MJ, MacLeod J, Evert A, et al. Academy of Nutrition and Dietetics Nutrition practice guideline for type 1 and type 2 diabetes in adults: Systematic review of evidence for medical nutrition therapy effectiveness and recommendations for integration into the nutrition care process. *J Acad Nutr Diet*. 2017;117(10):1659-1679.
24. Mazze RS, Strock ES, Simonson GD, Kendall DM, Cuddihy RM, Bergenstal RM. *Prevention, Detection, and Treatment of Diabetes in Adults Quick Guide*. 5th ed. St Louis Park, MN: International Diabetes Center at Park Nicollet; 2010.
25. Fox CS, Golden SH, Anderson C, et al. Update on prevention of cardiovascular disease in adults with type 2 diabetes mellitus in light of recent evidence: A scientific statement from the American Heart Association and the American Diabetes Association. *Circulation*. 2015;132(8):691-718.
26. Davidson P, Ross T, Castor C. Academy of Nutrition and Dietetics: Revised 2017 standards of practice and standards of professional performance for registered dietitian nutritionists (competent, proficient, and expert) in diabetes care. *J Acad Nutr Diet*. 2018;118(5):932-946. e48.
27. National Certification Board for Diabetes Educators. Certified Diabetes Educator eligibility requirements. https://www.ncbde.org/certification_info/eligibility-requirements/. Accessed March 7, 2018.
28. Minnesota Statute. 2018 Minnesota Statutes: 148.634 Prescription Protocol. www.revisor.mn.gov/statutes/cite/148.634. Accessed March 7, 2018.
29. Rickheim P, Flader J. *Type 2 Diabetes Basics*. 4th ed. St Louis Park, MN: International Diabetes Center at Park Nicollet; 2014.
30. Centers for Disease Control and Prevention (CDC). Behavioral Risk Factor Surveillance System Survey. Behavioral Risk Factor Surveillance System. https://www.cdc.gov/brfss/questionnaires/pdf-ques/2013-BRFSS_English.pdf. Published 2013. Accessed September 11, 2015.
31. Centers for Disease Control and Prevention (CDC). About adult BMI. https://www.cdc.gov/healthyweight/assessing/bmi/adult_bmi/index.html#InterpretedAdults. Accessed August 22, 2018.
32. Morisky DE, Green LW, Levine DM. Concurrent and predictive validity of a self-reported measure of medication adherence. *Med Care*. 1986;24(1):67-74.
33. Centers for Disease Control and Prevention (CDC). Behavioral Risk Factor Surveillance System Survey data. Behavioral Risk Factor Surveillance System. https://www.cdc.gov/brfss/questionnaires/pdf-ques/2009brfss.pdf. Published 2009. Accessed November 9, 2015.
34. American Heart Association. My Life Check™ Simple 7™ assessment. http://mylifecheck.heart.org/AssessmentTools2/main_en_US.html. Accessed October 26, 2015.
35. US Department of Health and Human Services and U.S. Department of Agriculture. *2015 — 2020 Dietary guidelines for Americans. 8th ed*. December 2015. https://health.gov/dietaryguidelines/2015/guidelines/. Accessed May 10, 2018.
36. Stanford Patient Education Research Center. Diabetes questionnaire. http://patienteducation.stanford.edu/research/breakfast.html. Accessed October 26, 2015.
37. Odegaard AO, Jacobs DR, Steffen LM, Van Horn L, Ludwig DS, Pereira MA. Breakfast frequency and development of metabolic risk. *Diabetes Care*. 2013;36(10):3100-3106.
38. Stata Statistical Software [computer program]. Release 15. College Station, TX: StataCorp LLC; 2017.
39. Academy Quality Management Committee. Academy of Nutrition and Dietetics: Revised 2017 scope of practice for the registered dietitian nutritionist. *J Acad Nutr Diet*. 2018;118(1):141-165.
40. Robinson JG, Conroy C, Wickemeyer WJ. A novel telephone-based system for management of secondary prevention to a low-density lipoprotein cholesterol >100 mg/dl. *Am J Cardiol*. 2000;85(3):305-308.
41. Worth JM, Davies RR, Durrington PN. A dietitian-led lipid clinic is effective. *Pract Diabetes Int*. 2006;23(5):221-226.
42. McEwen LN, Casagrande SS, Kuo S, Herman WH. Why are diabetes medications so expensive and what can be done to control their cost? *Curr Diab Rep*. 2017;17(9):71.
43. American Diabetes Association. Comprehensive medical evaluation and assessment of comorbidities: Standards of medical care in diabetes 2018. *Diabetes Care*. 2018;41:S28-S37.
44. Newman JD, Schwartzbard AZ, Weintraub HS, Goldberg IJ, Berger JS. Primary prevention of cardiovascular disease in diabetes mellitus. *J Am Coll Cardiol*. 2017;70(7):883-893.
45. Zgibor JC, Maloney MA, Malmi M Jr, et al. Effectiveness of certified diabetes educators following pre-approved protocols to redesign diabetes care delivery in primary care: Results of the REMEDIES 4D trial. *Contemp Clin Trials*. 2018;64(Jan):201-209.
46. Announcement. Community Preventive Services Task Force recommendation for team-based care for patients with type 2 diabetes. *Morbidity and Mortality Weekly Report (MMWR)*. https://www.cdc.gov/mmwr/volumes/66/wr/mm6623a6.htm. Published 2017. Accessed July 3, 2018.
47. Briggs Early K, Stanley K. Position of the Academy of Nutrition and Dietetics: The role of medical nutrition therapy and registered dietitian nutritionists in the prevention and treatment of prediabetes and type 2 diabetes. *J Acad Nutr Diet*. 2018;118(2):343-353.
48. Møller G, Keinke Andersen H, Snorgaard O. A systematic review and meta-analysis of nutrition therapy compared with dietary advice in patients with type 2 diabetes. *Am J Clin Nutr*. 2017;106(6):1394-1400.
49. Gourgari E, Wilhelm EE, Hassanzadeh H, Aroda VR, Shoulson I. A comprehensive review of the FDA-approved labels of diabetes drugs : Indications, safety, and emerging cardiovascular safety data. *J Diabetes Complications*. 2017;31(12):1719-1727.
50. American Diabetes Association. Cardiovascular disease and risk management. *Diabetes Care*. 2016;39(suppl 1):S60.
51. Centers for Disease Control and Prevention (CDC). Behavioral Risk Factor Surveillance System Questionnaire 2014. https://www.cdc.gov/brfss/questionnaires/pdf-ques/2014_BRFSS.pdf. Accessed October 26, 2015.

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STATEMENT OF POTENTIAL CONFLICT OF INTEREST

No potential conflict of interest was reported by the authors.

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AUTHOR CONTRIBUTIONS

G. A. Benson contributed to study design and drafting of the manuscript; J. Hayes contributed to study design and drafting of the manuscript; A. Sidebottom provided data analysis/interpretation of data; M. D. Miedema, contributed to study design and editing of manuscript; J. Boucher contributed to conception and editing of manuscript; M. Vacquier provided data analysis; A. Sillah provided data analysis; S. Gamam participated in manuscript drafting; J. J. VanWormer contributed to conception/design/interpretation of data and overall editing of manuscript.